

Selecting the Site of Andina Distribuciones' New CDC

by Alexander Garrido, Ph.D.

Learning Objectives

- Apply the weighted Center of Gravity (CoG) formula to locate a facility that minimizes weighted distance to demand.
- Explain why a single geometric centroid can be a poor location choice in a geographically dispersed network.
- Use K-Means clustering to segment demand into coherent regions, and understand the mathematical link between a weighted cluster centroid and the CoG formula.
- Build a simple cost trade-off model that balances transportation cost savings against the fixed cost of operating additional distribution centers.
- Translate a geometric (mathematical) recommendation into a realistic site recommendation, accounting for infrastructure constraints.

Business Context

Andina Distribuciones S.A.S. is a national consumer-goods distributor that currently ships all orders from a single warehouse located in Bogotá. As the company has grown into new regions of the country, the General Manager has started receiving complaints about long delivery times and rising freight costs to cities on the Caribbean coast and in the southwest of the country.

The General Manager has asked your team — the company's newly formed Logistics Analytics group — to evaluate whether the company should: (a) keep a single, centrally located CDC; or (b) move to a network of two or more regional CDCs. The decision should be based on quantitative analysis of where demand is concentrated, not on intuition alone.

You are given each city's approximate geographic coordinates and its average weekly demand in pallets. Your task is to recommend a network configuration (number and location of CDCs) that reduces total weighted transportation distance without letting fixed facility costs erase the savings.

The Data

Table 1 lists the 20 cities Andina Distribuciones currently serves, their approximate coordinates, and their average weekly demand.

<i>City</i>	<i>Latitude</i>	<i>Longitude</i>	<i>Weekly demand (pallets)</i>
Bogotá	4.7110	-74.0721	950
Medellín	6.2442	-75.5812	700
Cali	3.4516	-76.5320	620
Barranquilla	10.9639	-74.7964	480
Cartagena	10.3910	-75.4794	350
Bucaramanga	7.1193	-73.1227	300
Pereira	4.8087	-75.6906	220
Manizales	5.0689	-75.5174	180
Cúcuta	7.8939	-72.5078	260
Santa Marta	11.2408	-74.1990	210
Ibagué	4.4389	-75.2322	190
Villavicencio	4.1420	-73.6266	240
Pasto	1.2136	-77.2811	150
Montería	8.7479	-75.8814	170

Neiva	2.9273	-75.2819	160
Popayán	2.4448	-76.6147	130
Valledupar	10.4631	-73.2532	140
Armenia	4.5339	-75.6811	150
Sincelejo	9.3047	-75.3978	120
Tunja	5.5353	-73.3678	110

Table 1. Demand points for Andina Distribuciones S.A.S.

Methodology Overview

You will work through the problem in four stages, by using Python[®] programming ([Colab notebook](#)):

1. **Global Center of Gravity.** Compute a single weighted centroid for all 20 cities using the classic CoG formula and evaluate it as a single-CDC location.
2. **Clustering into regions.** Use [K-Means clustering](#) (weighted by demand) to group cities into 2–5 candidate regions, choosing the number of regions with the [elbow method](#).
3. **Weighted CoG per cluster.** Show that the weighted centroid of each cluster is mathematically the same CoG calculation as Stage 1, just applied within each region — this is the conceptual bridge between the two methods.
4. **Cost trade-off and real-site recommendation.** Compare total annual cost (transportation + fixed facility cost) across different numbers of CDCs, and convert each geometric centroid into a realistic recommendation by identifying the nearest real city with adequate infrastructure.

Key Formulas

Weighted Center of Gravity

For a set of demand points $i = 1, \dots, n$ with coordinates (x_i, y_i) and weight (demand) w_i , the CoG (X^*, Y^*) is:

$$X^* = \Sigma(w_i \cdot x_i) / \Sigma(w_i) \quad Y^* = \Sigma(w_i \cdot y_i) / \Sigma(w_i)$$

This is a simple approximation: it minimizes the sum of squared distances, not the true sum of distances. Note that when a K-Means cluster is fit with sample weight = demand, each resulting cluster center is exactly this same weighted average applied only to the points assigned to that cluster.

Great-circle (Haversine) distance

Because coordinates are latitude/longitude, straight-line Euclidean distance is not accurate over long distances. Python[®] ([notebook](#)) uses the [Haversine formula](#) to convert coordinate differences into kilometers before computing weighted distances and transportation costs.

Assignment Tasks

Work through the Python[®] ([notebook](#)) and answer the following in a short-written memo (max. 2 pages) addressed to the General Manager:

Part A — Report the global COG's coordinates and the total weighted distance for a single-CDC network. Comment on whether the exact geometric point is a realistic place to build a warehouse.

Part B — Report the [Elbow chart](#) and justify your chosen number of clusters (k). Explain, in your own words, why each cluster's weighted centroid is itself a CoG calculation.

Part C — Report the total annual cost (transport + fixed cost) for $k = 1$ through 5 CDCs, and state which k minimizes total cost under the assumptions given in Python[®] outbook (\$600,000/year fixed cost per CDC; \$0.05 per pallet-km).

Part D — For your recommended k , identify the nearest real city to each cluster centroid and propose it as the CDC site. Name at least one additional (non-distance) factor a real siting decision should consider (e.g., road access, port proximity, labor market, land cost, tax incentives).

Part E — Recommendation. In 3–5 sentences, tell the General Manager what to do and why, including one risk or limitation of your analysis.

Deliverables

- A completed copy of the Python code (share the link or download the .ipynb).
- A short-written memo (max. 2 pages) covering Parts A–E above.

Grading Rubric

<i>Criterion</i>	<i>What we're looking for</i>	<i>Points</i>
Global Center of Gravity (Part A)	Correct weighted-average formula, correct plot, sensible interpretation of the result (including its limitations).	15
Clustering & elbow method (Part B)	Elbow chart produced and read correctly; a justified choice of k with business reasoning, not just the mathematical minimum.	20
Weighted COG per cluster (Part B)	Correct use of <code>sample_weight</code> ; clear link drawn between K-Means centroids and the COG formula from Part A.	15
Cost trade-off analysis (Part C)	Correct total-cost function (transport + fixed costs); chart of cost vs. k ; correct identification of the cost-minimizing k .	20
Real-site translation (Part D)	Nearest real city identified per cluster; at least one qualitative siting factor discussed beyond distance.	15
Written recommendation & reflection	Clear recommendation to the General Manager; answers the reflection questions with reference to results, not just opinion.	15

Table 2. Grading rubric (100 points total).

Reflection Questions (for class discussion)

1. How does changing the fixed cost assumption (\$600,000/year per CDC) change the cost-minimizing number of facilities? What does that tell you about when a single-CDC vs. multi-CDC strategy makes sense?
2. This analysis only optimizes for cost. What supply chain resilience risks does a single, centralized CDC carry that a cost-only model does not capture?
3. What other variables (beyond location and demand volume) might change which cities cluster together in a real project?